Seminar Report of Team 6 on RIPE Atlas Traceroute Analysis for the Anchor 6360

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***Abstract*—Our project analyzes internet routing paths toward a regional destination using traceroute measurements from the RIPE Atlas platform. We focus on IPv4 traceroutes targeting Anchor 6360 in Salzburg, Austria (AS39878), collected from a wide range of global probes. The raw measurement data was processed using a custom Python pipeline that filters relevant traceroutes and enriches each hop with reverse DNS information, geolocation, and Autonomous System Numbers (ASNs). To ensure efficient processing, the enrichment was performed in a streaming manner and supported by a local caching mechanism to avoid redundant external lookups.**

We exported the enriched data in both hierarchical and tabular formats to support inspection and statistical analysis. Our results show that traffic toward the Austrian target frequently traverses international transit networks, with many intermediate hops located outside the destination country and a strong reliance on a small number of major transit providers.

To better understand and communicate these routing patterns, we visualize the enriched traceroute data using graph-based representations, where nodes correspond to ASNs and edges represent observed routing transitions. These visualizations highlight path diversity, routing convergence, and dominant transit networks, providing an intuitive view of complex internet routing behavior.

I. Dataset Description

A. Datasource and Collection

The dataset was obtained from RIPE Atlas [1], a global open internet measurement platform. The target selected for this analysis is Anchor 6360, located in Salzburg, Austria, managed by Salzburg AG.

Anchor FQDN: at-szg-as39878.anchors.atlas.ripe.net
Target IP: 91.135.161.162 (IPv4)
Target ASN: AS39878

We utilized Measurement ID 88302907 [2], which consists of periodic IPv4 traceroutes performed by varying RIPE Atlas probes toward the anchor. Probe measurements were selected over mesh measurements to analyze reachability from diverse consumer and data-center networks rather than just core-to-core infrastructure.

B. Dataset Structure

The raw input data (traceroute-2025-12-10T0000.bz2) is structured in JSON Lines (JSONL) format. We filtered this data using a custom parser to isolate our specific measurement ID. The resulting dataset consists of traceroute objects containing probe IDs, source and destination IP addresses and arrays of network hops.

II. Enriching Traceroute Dataset

A. Methodology and Tools

We selected Python [5] as programming language for the enrichment pipeline due to its support for JSON data manipulation and the availability of network analysis libraries. The core processing logic utilized the standard json library for parsing the line-delimited input and the csv library for structuring the final output. To interact with external enrichment services, we used the requests library [6], which simplifies HTTP interactions with RESTful APIs. A great challenge in processing network measurements is the latency associated with repeated API calls for frequently occurring IP addresses. To tackle this, we implemented a local caching mechanism in the script. By using Python dictionaries to store previously resolved IP addresses and probe metadata, our script ensures that no external API is queried twice for the same resource. This approach significantly reduced the execution time and ensured compliance with API rate limits. The script processes the input file in a streaming fashion, reading line-by-line, which prevents memory exhaustion and allows for the processing of arbitrarily large datasets.

B. Enrichment Methods and Theoretical Background

For every unique IP address identified in the traceroute path, we applied three distinct enrichment methods to improve the raw network data. The first method applied was Reverse DNS (rDNS) resolution. This process maps an IP address back to its associated domain name, which often contains semantic information embedded by network operators, such as the interface type, router function, and physical location. In our implementation, this is performed using the socket.gethostbyaddr function. Behind the scenes, this function triggers a DNS query for a PTR (Pointer) record within the in-addr.arpa domain hierarchy. While not all IP addresses have configured PTR records, successful lookups provide critical context regarding the network infrastructure.

The second enrichment method involved Geolocation, which maps an IP address to a specific country and city. This data is essential for visualizing the physical path of traffic across international borders. We utilized the ipinfo.io API [4] for this purpose. Behind the API interface, the service queries

a massive geolocation database that maps blocks of IP prefixes to geographic coordinates. This mapping is derived from data provided by Regional Internet Registries (RIRs) and internet topology mining. It is important to note that this method provides an estimate based on the registration of the IP block or the location of the routing infrastructure, rather than precise GPS coordinates.

The third method was Autonomous System Number (ASN) lookup. An ASN uniquely identifies the administrative organization, such as an Internet Service Provider (ISP) or a content delivery network, that controls a specific group of IP prefixes. Identifying the ASN allows for the analysis of inter-domain routing and the handoffs between different network operators. We performed this lookup using the RIPE Stat API [3]. Behind the scenes, this API queries global Border Gateway Protocol (BGP) routing tables to identify which Autonomous System is currently announcing the specific IP prefix to the global internet.

C. Dataset Structure and Schema

The enrichment pipeline produces two distinct output files to facilitate different modes of analysis. The first, enriched_traceroutes.json, retains the original hierarchical structure of the RIPE Atlas measurement but appends a detailed enrichment object to every hop in the trace. This format is good for deep inspection of specific paths. The second output, hop_enrichment.csv, is a flattened dataset designed for statistical analysis. The schema of this enriched CSV dataset includes the Measurement ID and Probe ID to identify the source of the trace. It records the specific Hop number and the IP address of the router at that stage. The enrichment fields include the resolved PTR hostname, the Autonomous System Number (ASN), the Country Code, as well as the estimated City. Finally, the Round-Trip Time (RTT) is recorded. This structured approach allows for the efficient aggregation of statistics regarding country distribution and ASN path diversity.

III. Traceroute Graphs

A. Visualization Approach and Methodology

To visualize the network topology targeting the Salzburg anchor, we adopted a graph-theoretical approach where the internet paths are modeled as a set of interconnected nodes and links. We iterated through the enriched traceroute dataset, treating every unique IP address found in the hops as a distinct node and every observed transition between consecutive hops as a directed edge. A directed graph structure was chosen over an undirected one to accurately represent the unidirectional flow of traceroute packets from the source probes toward the destination. The visualization was implemented using the Python programming language, utilizing the NetworkX library for graph construction and the Matplotlib library for rendering. To organize the complex topology into a readable format, we employed a force-directed layout algorithm. This method simulates physical forces between nodes, positioning high-degree hubs centrally while pushing less connected nodes, such as the source probes, to the periphery, effectively revealing the backbone structure of the network.

B. Objectives and Visual Analysis

The primary objective of this visualization was to spatially verify the ”funneling effect” suggested by our statistical analysis and to identify specific transit bottlenecks. By plotting the paths, we aimed to demonstrate how traffic from geographically diverse regions - such as the United States, Germany, and Czechia - merges into common upstream providers before reaching the final destination. The resulting graph illustrates a distinct converging tree topology. The outer periphery of the visualization consists of the distributed RIPE Atlas probes, while the core is dominated by a small number of high-degree nodes representing major transit providers. Specifically, the visualization confirms that AS201011 acts as the primary gateway for this anchor, with a dense cluster of edges converging onto its nodes before a single final link connects to the destination in AS39878 (Salzburg AG). This visual evidence underscores the reliance of the regional Austrian ISP on major international transit backbones for global reachability.

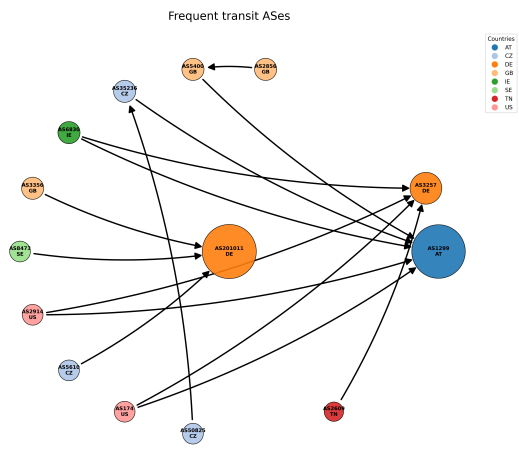

Fig. 1. Directed graph of traceroute paths targeting Anchor 6360. The topology shows diverse global probes converging into key transit ASNs before reaching the destination in Salzburg.

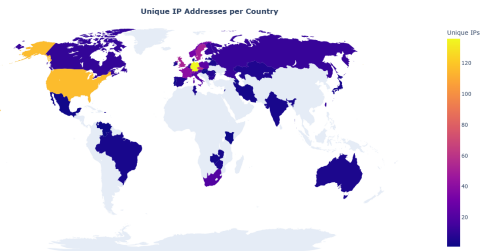

Fig. 2. Geographic heatmap showing the density of unique IP addresses encountered in traceroutes towards Anchor 6360. The map illustrates that while the destination is in Austria, the primary transit infrastructure traversed is located in Germany and the United States.

IV. Conclusion and Outlook

A. Summary of Findings and Project Reflections

This seminar report documented the successful development of a data processing pipeline capable of ingesting, filtering, and enriching raw network measurement data from the RIPE Atlas platform. By focusing on Anchor 6360 in Salzburg, Austria, we were able to reconstruct the network topology visible from a diverse set of global probes. The analysis of the enriched dataset yielded several significant insights regarding internet routing patterns. The most notable result was the geographic distribution of the intermediate hops; despite the target being located in Austria, our data revealed that a significant portion of traffic was routed through Germany (136 unique IPs) and the United States (116 unique IPs). This confirms that reachability to regional Austrian networks is heavily dependent on major international transit hubs in Frankfurt and North American backbones. Furthermore, the ASN analysis highlighted a strong consolidation of transit providers, with AS1299 appearing significantly more frequently (92 occurrences) than other Tier-1 providers, suggesting a potential single point of reliance for international traffic reaching this specific anchor.

In terms of project execution, the implementation of a local caching mechanism within the Python enrichment script was a decisive success. This approach allowed us to resolve 1,004 unique IP addresses without hitting API rate limits or suffering from excessive network latency. However, we encountered challenges regarding data volume. The initial raw dataset in BZ2 format required careful stream processing to avoid memory overflows, which dictated our decision to focus on a specific subset of probe measurements rather than the larger mesh dataset. An unforeseen surprise was the relatively low number of unique IPs located physically within Austria (18) compared to the number of probes and hops involved, underscoring how little of the "internet path" actually resides within the destination country.

B. Scalability and Future Directions

Based on our experience processing this sample, extending this work to analyze large-scale traceroute datasets, such as the full daily dumps from RIPE Atlas would require a fundamental shift in architecture. The current reliance on external REST APIs (ipinfo.io and RIPE Stat) for enrichment is the primary bottleneck; it is not scalable for millions of traceroutes due to network latency and strict rate limits. To handle a large-scale dataset, we would transition to using offline, downloadable databases such as MaxMind GeoLite2 [7] for geolocation and CAIDA's Routeviews [8] for ASN mapping. This would allow for local, low-latency lookups without external network calls. Additionally, the storage format would need to evolve from verbose JSON Lines to a compressed, binary format like Apache Parquet or a dedicated time-series database (e.g., ClickHouse [9]), which would significantly reduce storage costs and improve query performance for statistical analysis.

With the ability to process larger and more diverse datasets, we would propose conducting a longitudinal analysis of routing stability. By analyzing traceroutes over a longer period, we could detect "route flapping" events or shifts in transit providers that correlate with global internet outages or BGP hijacking attempts. Furthermore, a larger dataset would enable a "latency triangulation" study, where we could correlate the Round-Trip Time (RTT) increases across thousands of probes to pinpoint specific physical locations or Internet Exchange Points (IXPs) that are experiencing congestion. These additional analyses would be valuable for network operators seeking to optimize peering agreements and for researchers studying the resilience of the European internet infrastructure against localized failures.

Authorship Statement

The three of us followed the entire workflow together, but specialized in specific parts for the technical execution and report writing. We collectively performed the first step, which included choosing the anchor, recording its information, and downloading the appropriate traceroute dataset from RIPE Atlas. Subsequently, the work was distributed as follows: Mihajlo did Step 2, Jovana did Step 3, and Julian did Step 4. Regarding the written seminar report, every author wrote the section corresponding to their specific step, while the overview and outlook parts were done in union. Generative AI tools were used to assist in structuring the report layout and refining the English phrasing, but all code, data analysis, and results were produced by the authors.

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